

# Review on Interference and Diffraction

Phys2500J

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# Overview

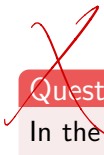
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**1. Problem Set**

**2. Diffraction**

# Mirage

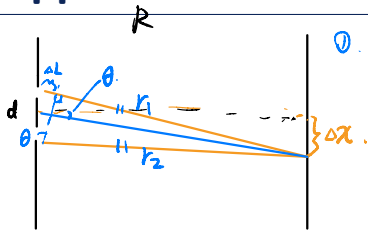
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## Question

In the desert, it is hot on the ground surface. As a result, the hot air near ground has smaller index than cooler air higher up. Assume the index has the relation with the height above the ground which is  $n(y) = 1 + \alpha y$ . Find the trajectory for the light rays which has an initial angle of  $30^\circ$  at the height of  $h$

# Approximation for Young's double



①.

$$\Delta L = d \sin \theta, \quad \theta \ll 1.$$

$$\sin \theta \approx \tan \theta = \frac{\Delta x}{R}.$$

$m$ th bright fringe

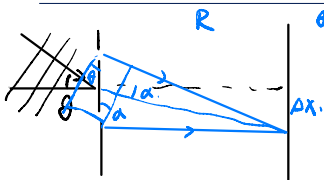
$$m\lambda = d \frac{\Delta x}{R} \Rightarrow \Delta x = m \frac{R}{d} \lambda.$$

②.  $r_1 = \sqrt{R^2 + \left(\frac{d}{2} + \Delta x\right)^2}$  ,  $R \gg d$ ;  $R \gg \Delta x$ ,  
 $\begin{cases} r_2 = \sqrt{R^2 + \left(\Delta x - \frac{d}{2}\right)^2} \end{cases}$

$$\Delta L = r_1 - r_2 \approx R \left( 1 + \frac{1}{2} \frac{\left(\frac{d}{2} + \Delta x\right)^2}{R^2} \right) - R \left( 1 + \frac{1}{2} \frac{\left(\Delta x - \frac{d}{2}\right)^2}{R^2} \right).$$

$$= \frac{d \Delta x}{R}.$$

# Two way of shifting



$\theta \ll 1$ .

$$d \sin \theta = d \sin \alpha$$

$$\sin \alpha \approx \frac{\Delta x}{R} = \sin \theta$$

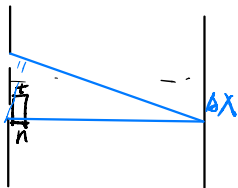
$$\Delta x = R \sin \theta$$

0th Brigh  $\Rightarrow$  Equal Optical length.

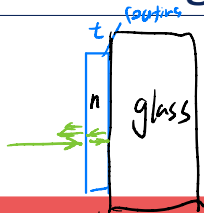
$$d \cdot \sin \theta = n \cdot t$$

$$\Rightarrow d \cdot \frac{\Delta x}{R} = nt$$

$$\Rightarrow \Delta x = \frac{ntR}{d}$$



# Anti-reflective coating



$$n_{\text{air}} < n_{\text{coating}} < n_{\text{glass}}.$$

## Question

Why the surface of your cellphone's camera lens looks colored? e.g. especially looks blue or purple.

$$n = 1.2.$$

$$\lambda_1 = 400 \text{ nm} \quad \text{transmitted,}$$

$$\lambda_2 = 620 \text{ nm} \quad \text{reflected,}$$

$$\Delta L = 2nt = (m + \frac{1}{2})\lambda_1, \quad \text{destructive for } \lambda_1,$$
$$= n\lambda_2. \quad \text{constructive for } \lambda_2.$$

# Diffraction

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## Definition

Light ray tends to bend around objects. This tendency is called diffraction. It is a property of waves.

## Examples

How to use Huygens's principle to analyze diffraction

- Fresnel diffraction: Source, screen, and obstacle are close together.
- Fraunhofer diffraction: Source, screen, and obstacle are far apart.

# Fraunhofer Diffraction

## Equation

Fraunhofer diffraction:

$$\tilde{U}(\theta) = C \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} T(x, y) e^{-ik \sin \theta (x \cos \varphi + y \sin \varphi)} dx dy$$

## Examples

Single slit:

Diagram: A vertical slit of width  $a$  is shown with a coordinate  $x$  along its length. An incident wave vector  $k$  is shown at an angle  $\theta$  to the normal.

$$U(\theta) = \int_{-\frac{a}{2}}^{\frac{a}{2}} e^{-ik \sin \theta \cdot x} dx$$

$$= \frac{1}{-ik \sin \theta} (e^{-ik \sin \theta \frac{a}{2}} - e^{ik \sin \theta \frac{a}{2}})$$

$$= \frac{\sin(k \sin \theta \frac{a}{2})}{k \sin \theta} = \frac{\sin(\frac{\beta}{2})}{\frac{\beta}{a}}$$

$$\beta = \left(\frac{2\pi}{\lambda}\right) \sin \theta \cdot a$$

$$I = I_0 \left( \frac{\sin(\frac{\beta}{2})}{\frac{\beta}{2}} \right)^2$$

$$\beta = \frac{2\pi}{\lambda} a \sin \theta$$

$$\sin \theta = \frac{m\lambda}{a}$$

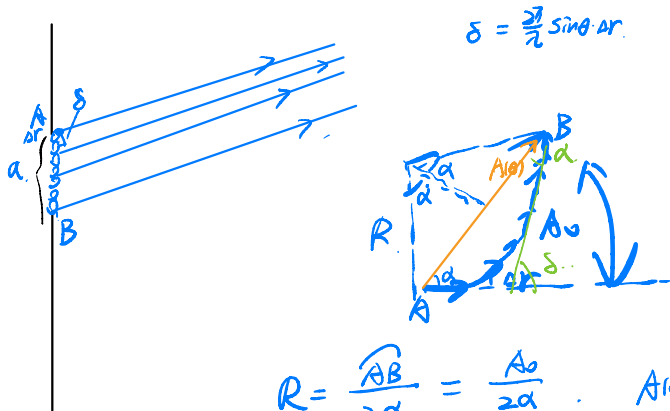
Diagram: A graph of intensity  $I$  versus  $\beta$  showing a central maximum and several smaller side lobes. The first zero is marked at  $\beta = \pi$ .

$\sin(\frac{\beta}{2}) = m\pi$

$\Downarrow$  zeros, mth dark fringes



# Fraunhofer Diffraction



$$\delta = \frac{2\pi}{\lambda} \sin\theta \cdot ar.$$

$$\delta_m = \frac{2\pi}{\lambda} \sin\theta \cdot a.$$

$$\alpha = \frac{\delta_m}{z} = \frac{\pi a}{\lambda} \sin\theta.$$

$$R = \frac{AB}{2\alpha} = \frac{A_0}{2\alpha} \quad A(\theta) = 2R \sin\alpha = \frac{A_0}{\alpha} \sin\alpha.$$

# Several Slits

## Equation

$$\begin{aligned}\tilde{U}(\theta) &= \int_{\Sigma} U_0(x) e^{ikr} dx \\ &= \sum e^{ikL_j} \int_{\Sigma_j} U_0(x_j) e^{-ikx_j \sin \theta} dx_j \\ &= \tilde{N}(\theta) \tilde{u}(\theta)\end{aligned}$$

*Handwritten notes:* A blue arrow points from the  $\sum$  term to the expression  $e^{ikL_0} / t$  written in blue ink above it.

$$r_j = L_j - x_j \sin \theta$$

$$L_j = L_1 + j \cdot (d \sin \theta)$$

Where

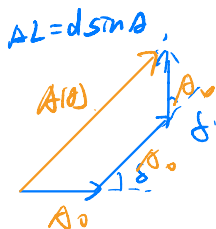
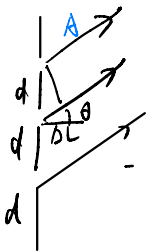
$$\tilde{N}(\theta) = C_N \frac{\sin N \frac{\delta}{2}}{\sin \frac{\delta}{2}}$$

$$\delta = \frac{2\pi d \sin \theta}{\lambda}$$

$$\tilde{u}(\theta) = C_u \frac{\sin \frac{\phi}{2}}{\frac{\phi}{2}}$$

$$\phi = \frac{2\pi a \sin \theta}{\lambda}$$

# Several Slits



$$\delta = \frac{2\pi d}{\lambda} \sin \theta$$

$$\beta = \frac{\delta}{2}$$

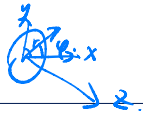
$$A(\theta) = A_0 + 2A \cos \delta$$

$$= A_0 \cdot \frac{\sin 3\beta}{\sin \beta}$$

By Sum up

$$\begin{aligned} \tilde{N}(\theta) &= e^{ikL_1} (1 + e^{ik\delta L} + e^{i2k\delta L} + \dots + e^{i(N-1)k\delta L}) \quad \beta = \frac{\pi d}{\lambda} \sin \theta \\ &= e^{ikL_1} \frac{1 - e^{i2N\beta}}{1 - e^{i2\beta}} = e^{ikL_1} e^{i(N-1)\beta} \left( \frac{e^{-iN\beta} - e^{iN\beta}}{e^{-i\beta} - e^{i\beta}} \right) \propto \frac{\sin N\beta}{\sin \beta} \end{aligned}$$

# Circular Apertures



$$\frac{d}{dx} (x^{\nu} J_{\nu}(x)) = x^{\nu} J_{\nu-1}(x)$$

first dark fringe:  $\sin \theta = 1.22 \frac{\lambda}{D} \rightarrow \text{Diameter.}$

$$U(\theta) = \int_0^{\frac{D}{2}} \int_0^{2\pi} e^{-ikr \sin \theta \cos \phi} r dr d\phi \quad \leftarrow \quad J_0(x) = \int_0^{\pi} \cos(\sin \theta) d\theta.$$

$$\propto \int_0^{\frac{D}{2}} r J_0(kr \sin \theta) \cdot dr$$

$$\frac{d}{dx} (x J_1(x)) = x J_0(x).$$

$$= \frac{1}{k \sin \theta} \cdot R \cdot J_1(kR \sin \theta).$$

first zero of  $J_1(x)$ .  $3.832 \approx \frac{2\pi}{\lambda} R \sin \theta = \frac{\pi D}{\lambda} \sin \theta$

$$\Rightarrow \sin \theta \approx 1.22 \frac{\lambda}{D}.$$

# Relation with Fourier Transform

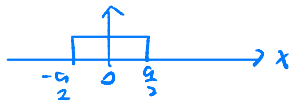
$$\int_{-\infty}^{\infty} T(x) e^{-ik \sin \theta x} dx$$

Young's double.

$$T(x) = \delta(x - \frac{d}{2}) + \delta(x + \frac{d}{2}), \quad \xrightarrow{F} \quad \cos(k \sin \theta \frac{d}{2}).$$



single slit.

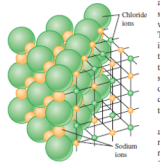
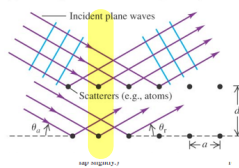


$$\begin{aligned} T(x) &= \text{rect}\left(\frac{x}{a}\right) \quad \xrightarrow{F} \quad \text{sinc}\left(k \sin \theta \frac{a}{2}\right). \\ &= u\left(x + \frac{a}{2}\right) - u\left(x - \frac{a}{2}\right). \end{aligned}$$

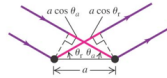
# X-ray diffraction

- The *Bragg condition* for constructive interference is

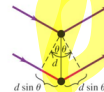
(a) Scattering of waves from a rectangular array



(b) Scattering from adjacent atoms in a row  
Interference from adjacent atoms in a row is constructive when the path lengths  $a \cos \theta_i$  and  $a \cos \theta_r$  are equal, so that the angle of incidence  $\theta_i$  equals the angle of reflection (scattering)  $\theta_r$ .



(c) Scattering from atoms in adjacent rows  
Interference from atoms in adjacent rows is constructive when the path difference  $2d \sin \theta$  is an integral number of wavelengths, as in Eq. (36.16).



$$2d \sin \theta = m\lambda.$$

**The End**